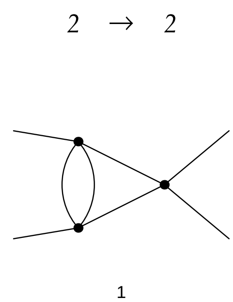
SK Matrix-Model Contributions for All L=2 Topologies

External assignment (by external-propagator appearance): $\phi_r^1(x_1)\,\phi_a^1(x_2)\,\phi_r^2(x_3)\,\phi_a^2(x_4)$, with $x_1 = (t, \mathbf{x})$, $x_2 = (0, \mathbf{0})$, $x_3 = (t, \mathbf{x})$, $x_4 = (0, \mathbf{0})$. Translation table used: same-fold $G_{rr} \to G_K = \frac{1}{2}(G^> + G^<)$, $G_{ra} \to G_R$, $G_{ar} \to G_A$, $G_{aa} \to 0$; cross-fold $G_{rr}^{12} \to G^<$, $G_{rr}^{21} \to G^>$, and any cross-fold propagator with an a endpoint vanishes. Contour ordering convention: contour (2) is later than contour (1), matching `contour_ra_translation.txt`. Each term carries explicit net matrix-size factor ${N_c}^{p-4}$ (raw loop count p minus the overall $1/N^4$ normalization).

Topology 1 (FeynArts Topology 2)

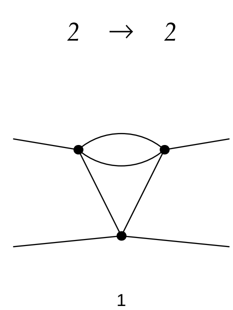


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 0. Net power in $C(t)$: -4. Explicit surviving terms:

$$\mathcal{T}_{1,1} = 128ig_2^3\,N_c^{-4}\,G^<(z_1,z_3)\;G^<(z_2,z_3)\;G_A(x_2,z_2)\;G_A(x_4,z_3)\;G_A(z_2,z_1)\;G_K(x_1,z_1)\;G_R(x_3,z_3)\;G_R(z_2,z_1)$$

$$\mathcal{T}_{1,2} = 128ig_2^3\,N_c^{-4}\,G^<(z_1,z_3)\;G^<(z_2,z_3)\;G_A(x_2,z_2)\;G_A(x_4,z_3)\;G_A(z_2,z_1)\;G_K(z_2,z_1)\;G_R(x_1,z_1)\;G_R(x_3,z_3)$$

Topology 2 (FeynArts Topology 2)

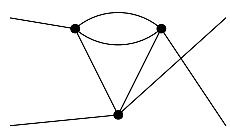


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 0. Net power in $C(t)$: -4.

No surviving SK contributions for this topology.

Topology 3 (FeynArts Topology 2)

2 $\rightarrow$ 2



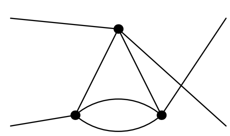
1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 0. Net power in $C(t)$: -4.

No surviving SK contributions for this topology.

Topology 4 (FeynArts Topology 2)

2 $\rightarrow$ 2

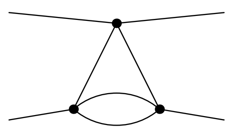


1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 0. Net power in $C(t)$: -4.

No surviving SK contributions for this topology.

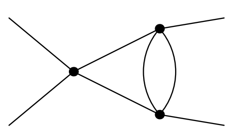
Topology 5 (FeynArts Topology 2)



Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 0. Net power in $C(t)$: -4.

No surviving SK contributions for this topology.

Topology 6 (FeynArts Topology 2)



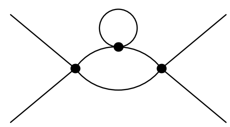
Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 0. Net power in $C(t)$: -4. Explicit surviving terms:

$$\mathcal{T}_{6,1} = 128 i g_2^3 \, N_c^{-4} \, G^{>}(z_2, z_1) \; G^{>}(z_3, z_1) \; G_A(x_2, z_1) \; G_A(x_4, z_3) \; G_A(z_3, z_2) \; G_K(x_3, z_2) \; G_R(x_1, z_1) \; G_R(z_3, z_2)$$

$$\mathcal{T}_{6,2} = 128 i g_2^3 \, N_c^{-4} \, G^{>}(z_2, z_1) \; G^{>}(z_3, z_1) \; G_A(x_2, z_1) \; G_A(x_4, z_3) \; G_A(z_3, z_2) \; G_K(z_3, z_2) \; G_R(x_1, z_1) \; G_R(x_3, z_2)$$

Topology 7 (FeynArts Topology 2)

$2 \;\to\; 2$



1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\mathcal{T}_{7,1} = 64ig_2^3\,N_c^{-3}\,G^<(z_1,z_2)\;G^<(z_1,z_3)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_A(z_2,z_3)\;G_A(z_3,z_3)\;G_K(x_3,z_2)\;G_R(x_1,z_1)$$

$$\mathcal{T}_{7,2} = 64ig_2^3\,N_c^{-3}\,G^<(z_1,z_2)\;G^<(z_1,z_3)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_A(z_2,z_3)\;G_K(x_3,z_2)\;G_R(x_1,z_1)\;G_R(z_3,z_3)$$

$$\mathcal{T}_{7,3} = 64ig_2^3\,N_c^{-3}\,G^<(z_1,z_2)\;G^<(z_1,z_3)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_A(z_3,z_3)\;G_K(z_2,z_3)\;G_R(x_1,z_1)\;G_R(x_3,z_2)$$

$$\mathcal{T}_{7,4} = 64ig_2^3\,N_c^{-3}\,G^<(z_1,z_2)\;G^<(z_1,z_3)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_K(z_2,z_3)\;G_R(x_1,z_1)\;G_R(x_3,z_2)\;G_R(z_3,z_3)$$

$$\mathcal{T}_{7,5} = 64ig_2^3\,N_c^{-3}\,G^<(z_1,z_2)\;G^<(z_1,z_3)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_K(z_3,z_3)\;G_R(x_1,z_1)\;G_R(x_3,z_2)\;G_R(z_2,z_3)$$

$$\mathcal{T}_{7,6} = 64ig_2^3\,N_c^{-3}\,G^<(z_1,z_2)\;G^>(z_2,z_3)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_A(z_1,z_3)\;G_A(z_3,z_3)\;G_K(x_1,z_1)\;G_R(x_3,z_2)$$

$$\mathcal{T}_{7,7} = 64ig_2^3\,N_c^{-3}\,G^<(z_1,z_2)\;G^>(z_2,z_3)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_A(z_1,z_3)\;G_K(x_1,z_1)\;G_R(x_3,z_2)\;G_R(z_3,z_3)$$

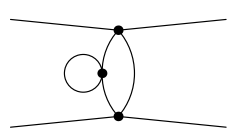
$$\mathcal{T}_{7,8} = 64ig_2^3\,N_c^{-3}\,G^<(z_1,z_2)\;G^>(z_2,z_3)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_A(z_3,z_3)\;G_K(z_1,z_3)\;G_R(x_1,z_1)\;G_R(x_3,z_2)$$

$$\mathcal{T}_{7,9} = 64ig_2^3\,N_c^{-3}\,G^<(z_1,z_2)\;G^>(z_2,z_3)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_K(z_1,z_3)\;G_R(x_1,z_1)\;G_R(x_3,z_2)\;G_R(z_3,z_3)$$

$$\mathcal{T}_{7,10} = 64ig_2^3\,N_c^{-3}\,G^<(z_1,z_2)\;G^>(z_2,z_3)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_K(z_3,z_3)\;G_R(x_1,z_1)\;G_R(x_3,z_2)\;G_R(z_1,z_3)$$

Topology 8 (FeynArts Topology 2)

$2 \;\to\; 2$



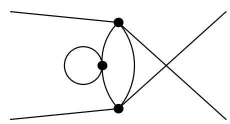
1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3.

No surviving SK contributions for this topology.

Topology 9 (FeynArts Topology 2)

$2 \;\rightarrow\; 2$



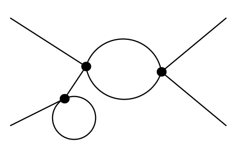
1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3.

No surviving SK contributions for this topology.

Topology 10 (FeynArts Topology 4)

$2 \;\rightarrow\; 2$



1

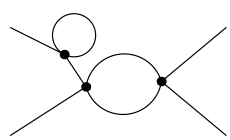
Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\mathcal{T}_{10,1} = 128 i g_2^3\, N_c^{-3}\, G^{<}(x_1, z_1)\; G^{<}(z_2, z_1)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_A(z_1, z_3)\; G_A(z_2, z_2)\; G_K(x_3, z_3)\; G_R(z_1, z_3)$$

$$\mathcal{T}_{10,2} = 128 i g_2^3\, N_c^{-3}\, G^<(x_1,z_1)\; G^<(z_2,z_1)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_1,z_3)\; G_A(z_2,z_2)\; G_K(z_1,z_3)\; G_R(x_3,z_3)$$
$$\mathcal{T}_{10,3} = 128 i g_2^3\, N_c^{-3}\, G^<(x_1,z_1)\; G^<(z_2,z_1)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_1,z_3)\; G_K(x_3,z_3)\; G_R(z_1,z_3)\; G_R(z_2,z_2)$$
$$\mathcal{T}_{10,4} = 128 i g_2^3\, N_c^{-3}\, G^<(x_1,z_1)\; G^<(z_2,z_1)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_1,z_3)\; G_K(z_1,z_3)\; G_R(x_3,z_3)\; G_R(z_2,z_2)$$
$$\mathcal{T}_{10,5} = 64 i g_2^3\, N_c^{-3}\, G^<(z_1,z_3)\; G^<(z_1,z_3)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_2,z_1)\; G_K(z_2,z_2)\; G_R(x_1,z_1)\; G_R(x_3,z_3)$$
$$\mathcal{T}_{10,6} = 64 i g_2^3\, N_c^{-3}\, G^<(z_1,z_3)\; G^<(z_1,z_3)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_2,z_2)\; G_K(x_1,z_1)\; G_R(x_3,z_3)\; G_R(z_2,z_1)$$
$$\mathcal{T}_{10,7} = 64 i g_2^3\, N_c^{-3}\, G^<(z_1,z_3)\; G^<(z_1,z_3)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_2,z_2)\; G_K(z_2,z_1)\; G_R(x_1,z_1)\; G_R(x_3,z_3)$$
$$\mathcal{T}_{10,8} = 64 i g_2^3\, N_c^{-3}\, G^<(z_1,z_3)\; G^<(z_1,z_3)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_K(x_1,z_1)\; G_R(x_3,z_3)\; G_R(z_2,z_1)\; G_R(z_2,z_2)$$
$$\mathcal{T}_{10,9} = 64 i g_2^3\, N_c^{-3}\, G^<(z_1,z_3)\; G^<(z_1,z_3)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_K(z_2,z_1)\; G_R(x_1,z_1)\; G_R(x_3,z_3)\; G_R(z_2,z_2)$$

Topology 11 (FeynArts Topology 4)

$2 \rightarrow 2$



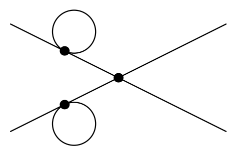
1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\mathcal{T}_{11,1} = 64 i g_2^3\, N_c^{-3}\, G^<(z_2,z_3)\; G^<(z_2,z_3)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_1,z_1)\; G_A(z_2,z_1)\; G_K(x_1,z_1)\; G_R(x_3,z_3)$$
$$\mathcal{T}_{11,2} = 64 i g_2^3\, N_c^{-3}\, G^<(z_2,z_3)\; G^<(z_2,z_3)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_2,z_1)\; G_K(x_1,z_1)\; G_R(x_3,z_3)\; G_R(z_1,z_1)$$
$$\mathcal{T}_{11,3} = 64 i g_2^3\, N_c^{-3}\, G^<(z_2,z_3)\; G^<(z_2,z_3)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_2,z_1)\; G_K(z_1,z_1)\; G_R(x_1,z_1)\; G_R(x_3,z_3)$$

Topology 12 (FeynArts Topology 4)

2 → 2



1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 2. Net power in $C(t)$: -2. Explicit surviving terms:

$$\mathcal{T}_{12,1} = 64ig_2^3\, N_c^{-2}\, G^<(x_1, z_1)\; G^<(z_2, z_3)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_A(z_1, z_1)\; G_A(z_2, z_2)\; G_K(x_3, z_3)\; G_R(z_1, z_3)$$

$$\mathcal{T}_{12,2} = 64ig_2^3\, N_c^{-2}\, G^<(x_1, z_1)\; G^<(z_2, z_3)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_A(z_1, z_1)\; G_A(z_2, z_2)\; G_K(z_1, z_3)\; G_R(x_3, z_3)$$

$$\mathcal{T}_{12,3} = 64ig_2^3\, N_c^{-2}\, G^<(x_1, z_1)\; G^<(z_2, z_3)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_A(z_1, z_1)\; G_K(x_3, z_3)\; G_R(z_1, z_3)\; G_R(z_2, z_2)$$

$$\mathcal{T}_{12,4} = 64ig_2^3\, N_c^{-2}\, G^<(x_1, z_1)\; G^<(z_2, z_3)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_A(z_1, z_1)\; G_K(z_1, z_3)\; G_R(x_3, z_3)\; G_R(z_2, z_2)$$

$$\mathcal{T}_{12,5} = 64ig_2^3\, N_c^{-2}\, G^<(x_1, z_1)\; G^<(z_2, z_3)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_A(z_1, z_3)\; G_A(z_2, z_2)\; G_K(z_1, z_1)\; G_R(x_3, z_3)$$

$$\mathcal{T}_{12,6} = 64ig_2^3\, N_c^{-2}\, G^<(x_1, z_1)\; G^<(z_2, z_3)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_A(z_1, z_3)\; G_K(z_1, z_1)\; G_R(x_3, z_3)\; G_R(z_2, z_2)$$

$$\mathcal{T}_{12,7} = 64ig_2^3\, N_c^{-2}\, G^<(x_1, z_1)\; G^<(z_2, z_3)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_A(z_2, z_2)\; G_K(x_3, z_3)\; G_R(z_1, z_1)\; G_R(z_1, z_3)$$

$$\mathcal{T}_{12,8} = 64ig_2^3\, N_c^{-2}\, G^<(x_1, z_1)\; G^<(z_2, z_3)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_A(z_2, z_2)\; G_K(z_1, z_3)\; G_R(x_3, z_3)\; G_R(z_1, z_1)$$

$$\mathcal{T}_{12,9} = 64ig_2^3\, N_c^{-2}\, G^<(x_1, z_1)\; G^<(z_2, z_3)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_K(x_3, z_3)\; G_R(z_1, z_1)\; G_R(z_1, z_3)\; G_R(z_2, z_2)$$

$$\mathcal{T}_{12,10} = 64ig_2^3\, N_c^{-2}\, G^<(x_1, z_1)\; G^<(z_2, z_3)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_K(z_1, z_3)\; G_R(x_3, z_3)\; G_R(z_1, z_1)\; G_R(z_2, z_2)$$

$$\mathcal{T}_{12,11} = 64ig_2^3\, N_c^{-2}\, G^<(z_1, z_3)\; G^<(z_2, z_3)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_A(z_1, z_1)\; G_A(z_2, z_2)\; G_K(x_1, z_1)\; G_R(x_3, z_3)$$

$$\mathcal{T}_{12,12} = 64ig_2^3\, N_c^{-2}\, G^<(z_1, z_3)\; G^<(z_2, z_3)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_A(z_1, z_1)\; G_K(x_1, z_1)\; G_R(x_3, z_3)\; G_R(z_2, z_2)$$

$$\mathcal{T}_{12,13} = 64ig_2^3\, N_c^{-2}\, G^<(z_1, z_3)\; G^<(z_2, z_3)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_A(z_2, z_2)\; G_K(x_1, z_1)\; G_R(x_3, z_3)\; G_R(z_1, z_1)$$

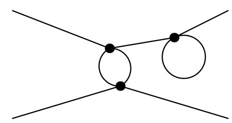
$$\mathcal{T}_{12,14} = 64ig_2^3\, N_c^{-2}\, G^<(z_1, z_3)\; G^<(z_2, z_3)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_A(z_2, z_2)\; G_K(z_1, z_1)\; G_R(x_1, z_1)\; G_R(x_3, z_3)$$

$$\mathcal{T}_{12,15} = 64ig_2^3\, N_c^{-2}\, G^<(z_1, z_3)\; G^<(z_2, z_3)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_K(x_1, z_1)\; G_R(x_3, z_3)\; G_R(z_1, z_1)\; G_R(z_2, z_2)$$

$$\mathcal{T}_{12,16} = 64ig_2^3\, N_c^{-2}\, G^<(z_1, z_3)\; G^<(z_2, z_3)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_K(z_1, z_1)\; G_R(x_1, z_1)\; G_R(x_3, z_3)\; G_R(z_2, z_2)$$

Topology 13 (FeynArts Topology 4)

$2 \rightarrow 2$



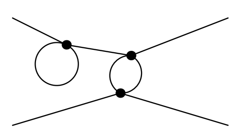
1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3.

No surviving SK contributions for this topology.

Topology 14 (FeynArts Topology 4)

$2 \rightarrow 2$

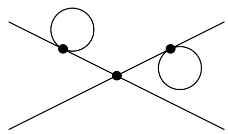


1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3.

No surviving SK contributions for this topology.

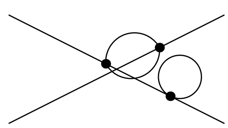
Topology 15 (FeynArts Topology 4)



Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 2. Net power in $C(t)$: -2.

No surviving SK contributions for this topology.

Topology 16 (FeynArts Topology 4)



Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\mathcal{T}_{16,1} = 128ig_2^3\,N_c^{-3}\,G^>(x_3,z_2)\;G^>(z_3,z_1)\;G_A(x_2,z_2)\;G_A(x_4,z_3)\;G_A(z_1,z_2)\;G_A(z_3,z_3)\;G_K(x_1,z_1)\;G_R(z_1,z_2)$$

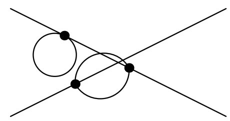
$$\mathcal{T}_{16,2} = 128ig_2^3\,N_c^{-3}\,G^>(x_3,z_2)\;G^>(z_3,z_1)\;G_A(x_2,z_2)\;G_A(x_4,z_3)\;G_A(z_1,z_2)\;G_K(x_1,z_1)\;G_R(z_1,z_2)\;G_R(z_3,z_3)$$

$$\mathcal{T}_{16,3} = 128ig_2^3\,N_c^{-3}\,G^>(x_3,z_2)\;G^>(z_3,z_1)\;G_A(x_2,z_2)\;G_A(x_4,z_3)\;G_A(z_3,z_3)\;G_K(z_1,z_2)\;G_R(x_1,z_1)\;G_R(z_1,z_2)$$

$$\mathcal{T}_{16,4} = 128ig_2^3\,N_c^{-3}\,G^>(x_3,z_2)\;G^>(z_3,z_1)\;G_A(x_2,z_2)\;G_A(x_4,z_3)\;G_K(z_1,z_2)\;G_R(x_1,z_1)\;G_R(z_1,z_2)\;G_R(z_3,z_3)$$

Topology 17 (FeynArts Topology 4)

$2 \rightarrow 2$



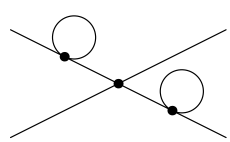
1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3.

No surviving SK contributions for this topology.

Topology 18 (FeynArts Topology 4)

$2 \rightarrow 2$



1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 2. Net power in $C(t)$: -2. Explicit surviving terms:

$$\mathcal{T}_{18,1} = 64ig_2^3\,N_c^{-2}\,G^{>}(x_3,z_2)\;G^{>}(z_3,z_2)\;G_A(x_2,z_2)\;G_A(x_4,z_3)\;G_A(z_1,z_1)\;G_A(z_3,z_3)\;G_K(x_1,z_1)\;G_R(z_1,z_2)$$

$$\mathcal{T}_{18,2} = 64ig_2^3\,N_c^{-2}\,G^{>}(x_3,z_2)\;G^{>}(z_3,z_2)\;G_A(x_2,z_2)\;G_A(x_4,z_3)\;G_A(z_1,z_1)\;G_K(x_1,z_1)\;G_R(z_1,z_2)\;G_R(z_3,z_3)$$

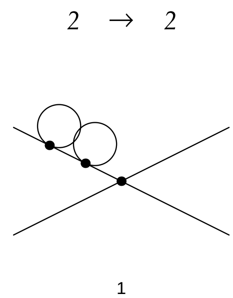
$$\mathcal{T}_{18,3} = 64ig_2^3\,N_c^{-2}\,G^{>}(x_3,z_2)\;G^{>}(z_3,z_2)\;G_A(x_2,z_2)\;G_A(x_4,z_3)\;G_A(z_3,z_3)\;G_K(x_1,z_1)\;G_R(z_1,z_1)\;G_R(z_1,z_2)$$

$$\mathcal{T}_{18,4} = 64ig_2^3\,N_c^{-2}\,G^{>}(x_3,z_2)\;G^{>}(z_3,z_2)\;G_A(x_2,z_2)\;G_A(x_4,z_3)\;G_A(z_3,z_3)\;G_K(z_1,z_1)\;G_R(x_1,z_1)\;G_R(z_1,z_2)$$

$$\mathcal{T}_{18,5} = 64ig_2^3\,N_c^{-2}\,G^{>}(x_3,z_2)\;G^{>}(z_3,z_2)\;G_A(x_2,z_2)\;G_A(x_4,z_3)\;G_K(x_1,z_1)\;G_R(z_1,z_1)\;G_R(z_1,z_2)\;G_R(z_3,z_3)$$

$$\mathcal{T}_{18,6} = 64ig_2^3\,N_c^{-2}\,G^{>}(x_3,z_2)\;G^{>}(z_3,z_2)\;G_A(x_2,z_2)\;G_A(x_4,z_3)\;G_K(z_1,z_1)\;G_R(x_1,z_1)\;G_R(z_1,z_2)\;G_R(z_3,z_3)$$

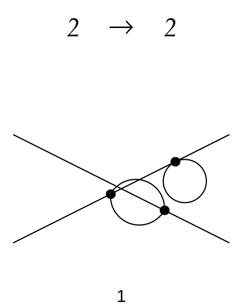
Topology 19 (FeynArts Topology 4)



Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 2. Net power in $C(t)$: -2.

No surviving SK contributions for this topology.

Topology 20 (FeynArts Topology 4)

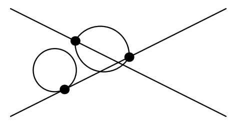


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3.

No surviving SK contributions for this topology.

Topology 21 (FeynArts Topology 4)

2 → 2



1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\mathcal{T}_{21,1} = 128ig_2^3\,N_c^{-3}\,G^{<}\left(x_1,z_1\right)\,G^{>}\left(z_3,z_2\right)\,G_A\left(x_2,z_2\right)\,G_A\left(x_4,z_1\right)\,G_A\left(z_2,z_2\right)\,G_A\left(z_3,z_1\right)\,G_K\left(x_3,z_3\right)\,G_R\left(z_3,z_1\right)$$

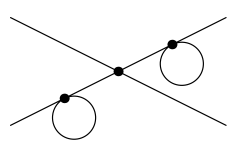
$$\mathcal{T}_{21,2} = 128ig_2^3\,N_c^{-3}\,G^{<}\left(x_1,z_1\right)\,G^{>}\left(z_3,z_2\right)\,G_A\left(x_2,z_2\right)\,G_A\left(x_4,z_1\right)\,G_A\left(z_2,z_2\right)\,G_K\left(z_3,z_1\right)\,G_R\left(x_3,z_3\right)\,G_R\left(z_3,z_1\right)$$

$$\mathcal{T}_{21,3} = 128ig_2^3\,N_c^{-3}\,G^{<}\left(x_1,z_1\right)\,G^{>}\left(z_3,z_2\right)\,G_A\left(x_2,z_2\right)\,G_A\left(x_4,z_1\right)\,G_A\left(z_3,z_1\right)\,G_K\left(x_3,z_3\right)\,G_R\left(z_2,z_2\right)\,G_R\left(z_3,z_1\right)$$

$$\mathcal{T}_{21,4} = 128ig_2^3\,N_c^{-3}\,G^{<}\left(x_1,z_1\right)\,G^{>}\left(z_3,z_2\right)\,G_A\left(x_2,z_2\right)\,G_A\left(x_4,z_1\right)\,G_K\left(z_3,z_1\right)\,G_R\left(x_3,z_3\right)\,G_R\left(z_2,z_2\right)\,G_R\left(z_3,z_1\right)$$

Topology 22 (FeynArts Topology 4)

2 → 2



1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 2. Net power in $C(t)$: -2. Explicit surviving terms:

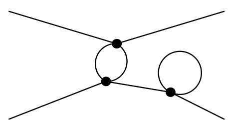
$$\mathcal{T}_{22,1} = 64ig_2^3\,N_c^{-2}\,G^{<}\left(x_1,z_1\right)\,G^{<}\left(z_2,z_1\right)\,G_A\left(x_2,z_2\right)\,G_A\left(x_4,z_1\right)\,G_A\left(z_2,z_2\right)\,G_A\left(z_3,z_3\right)\,G_K\left(x_3,z_3\right)\,G_R\left(z_3,z_1\right)$$

$$\mathcal{T}_{22,2} = 64ig_2^3\,N_c^{-2}\,G^{<}\left(x_1,z_1\right)\,G^{<}\left(z_2,z_1\right)\,G_A\left(x_2,z_2\right)\,G_A\left(x_4,z_1\right)\,G_A\left(z_2,z_2\right)\,G_K\left(x_3,z_3\right)\,G_R\left(z_3,z_1\right)\,G_R\left(z_3,z_3\right)$$

$$\mathcal{T}_{22,3} = 64ig_2^3\,N_c^{-2}\,G^<(x_1,z_1)\;G^<(z_2,z_1)\;G_A(x_2,z_2)\;G_A(x_4,z_1)\;G_A(z_2,z_2)\;G_K(z_3,z_3)\;G_R(x_3,z_3)\;G_R(z_3,z_1)$$
$$\mathcal{T}_{22,4} = 64ig_2^3\,N_c^{-2}\,G^<(x_1,z_1)\;G^<(z_2,z_1)\;G_A(x_2,z_2)\;G_A(x_4,z_1)\;G_A(z_3,z_3)\;G_K(x_3,z_3)\;G_R(z_2,z_2)\;G_R(z_3,z_1)$$
$$\mathcal{T}_{22,5} = 64ig_2^3\,N_c^{-2}\,G^<(x_1,z_1)\;G^<(z_2,z_1)\;G_A(x_2,z_2)\;G_A(x_4,z_1)\;G_K(x_3,z_3)\;G_R(z_2,z_2)\;G_R(z_3,z_1)\;G_R(z_3,z_3)$$
$$\mathcal{T}_{22,6} = 64ig_2^3\,N_c^{-2}\,G^<(x_1,z_1)\;G^<(z_2,z_1)\;G_A(x_2,z_2)\;G_A(x_4,z_1)\;G_K(z_3,z_3)\;G_R(x_3,z_3)\;G_R(z_2,z_2)\;G_R(z_3,z_1)$$

Topology 23 (FeynArts Topology 4)

2 → 2



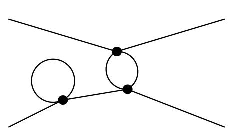
1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\mathcal{T}_{23,1} = 128ig_2^3\,N_c^{-3}\,G^>(x_3,z_1)\;G^>(z_3,z_2)\;G_A(x_2,z_2)\;G_A(x_4,z_3)\;G_A(z_2,z_1)\;G_A(z_3,z_3)\;G_K(x_1,z_1)\;G_R(z_2,z_1)$$
$$\mathcal{T}_{23,2} = 128ig_2^3\,N_c^{-3}\,G^>(x_3,z_1)\;G^>(z_3,z_2)\;G_A(x_2,z_2)\;G_A(x_4,z_3)\;G_A(z_2,z_1)\;G_A(z_3,z_3)\;G_K(z_2,z_1)\;G_R(x_1,z_1)$$
$$\mathcal{T}_{23,3} = 128ig_2^3\,N_c^{-3}\,G^>(x_3,z_1)\;G^>(z_3,z_2)\;G_A(x_2,z_2)\;G_A(x_4,z_3)\;G_A(z_2,z_1)\;G_K(x_1,z_1)\;G_R(z_2,z_1)\;G_R(z_3,z_3)$$
$$\mathcal{T}_{23,4} = 128ig_2^3\,N_c^{-3}\,G^>(x_3,z_1)\;G^>(z_3,z_2)\;G_A(x_2,z_2)\;G_A(x_4,z_3)\;G_A(z_2,z_1)\;G_K(z_2,z_1)\;G_R(x_1,z_1)\;G_R(z_3,z_3)$$

Topology 24 (FeynArts Topology 4)

2 → 2



1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\mathcal{T}_{24,1} = 128ig_2^3\, N_c^{-3}\, G^<(x_1,z_1)\; G^>(z_3,z_2)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_2,z_2)\; G_A(z_3,z_1)\; G_K(x_3,z_1)\; G_R(z_3,z_1)$$

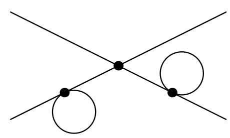
$$\mathcal{T}_{24,2} = 128ig_2^3\, N_c^{-3}\, G^<(x_1,z_1)\; G^>(z_3,z_2)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_2,z_2)\; G_A(z_3,z_1)\; G_K(z_3,z_1)\; G_R(x_3,z_1)$$

$$\mathcal{T}_{24,3} = 128ig_2^3\, N_c^{-3}\, G^<(x_1,z_1)\; G^>(z_3,z_2)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_3,z_1)\; G_K(x_3,z_1)\; G_R(z_2,z_2)\; G_R(z_3,z_1)$$

$$\mathcal{T}_{24,4} = 128ig_2^3\, N_c^{-3}\, G^<(x_1,z_1)\; G^>(z_3,z_2)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_3,z_1)\; G_K(z_3,z_1)\; G_R(x_3,z_1)\; G_R(z_2,z_2)$$

Topology 25 (FeynArts Topology 4)

$2 \rightarrow 2$



1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 2. Net power in $C(t)$: -2. Explicit surviving terms:

$$\mathcal{T}_{25,1} = 64ig_2^3\, N_c^{-2}\, G^<(x_1,z_1)\; G^<(z_2,z_1)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_2,z_2)\; G_A(z_3,z_1)\; G_K(z_3,z_3)\; G_R(x_3,z_1)$$

$$\mathcal{T}_{25,2} = 64ig_2^3\, N_c^{-2}\, G^<(x_1,z_1)\; G^<(z_2,z_1)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_2,z_2)\; G_A(z_3,z_3)\; G_K(x_3,z_1)\; G_R(z_3,z_1)$$

$$\mathcal{T}_{25,3} = 64ig_2^3\, N_c^{-2}\, G^<(x_1,z_1)\; G^<(z_2,z_1)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_2,z_2)\; G_A(z_3,z_3)\; G_K(z_3,z_1)\; G_R(x_3,z_1)$$

$$\mathcal{T}_{25,4} = 64ig_2^3\, N_c^{-2}\, G^<(x_1,z_1)\; G^<(z_2,z_1)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_2,z_2)\; G_K(x_3,z_1)\; G_R(z_3,z_1)\; G_R(z_3,z_3)$$

$$\mathcal{T}_{25,5} = 64ig_2^3\, N_c^{-2}\, G^<(x_1,z_1)\; G^<(z_2,z_1)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_2,z_2)\; G_K(z_3,z_1)\; G_R(x_3,z_1)\; G_R(z_3,z_3)$$

$$\mathcal{T}_{25,6} = 64ig_2^3\, N_c^{-2}\, G^<(x_1,z_1)\; G^<(z_2,z_1)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_3,z_1)\; G_K(z_3,z_3)\; G_R(x_3,z_1)\; G_R(z_2,z_2)$$

$$\mathcal{T}_{25,7} = 64ig_2^3\, N_c^{-2}\, G^<(x_1,z_1)\; G^<(z_2,z_1)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_3,z_3)\; G_K(x_3,z_1)\; G_R(z_2,z_2)\; G_R(z_3,z_1)$$

$$\mathcal{T}_{25,8} = 64ig_2^3\, N_c^{-2}\, G^<(x_1,z_1)\; G^<(z_2,z_1)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_3,z_3)\; G_K(z_3,z_1)\; G_R(x_3,z_1)\; G_R(z_2,z_2)$$

$$\mathcal{T}_{25,9} = 64ig_2^3\, N_c^{-2}\, G^<(x_1,z_1)\; G^<(z_2,z_1)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_K(x_3,z_1)\; G_R(z_2,z_2)\; G_R(z_3,z_1)\; G_R(z_3,z_3)$$

$$\mathcal{T}_{25,10} = 64ig_2^3\, N_c^{-2}\, G^<(x_1,z_1)\; G^<(z_2,z_1)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_K(z_3,z_1)\; G_R(x_3,z_1)\; G_R(z_2,z_2)\; G_R(z_3,z_3)$$

$$\mathcal{T}_{25,11} = 64ig_2^3\, N_c^{-2}\, G^>(x_3,z_1)\; G^>(z_3,z_1)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_2,z_1)\; G_A(z_3,z_3)\; G_K(z_2,z_2)\; G_R(x_1,z_1)$$

$$\mathcal{T}_{25,12} = 64ig_2^3\, N_c^{-2}\, G^>(x_3,z_1)\; G^>(z_3,z_1)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_2,z_1)\; G_K(z_2,z_2)\; G_R(x_1,z_1)\; G_R(z_3,z_3)$$

$$\mathcal{T}_{25,13} = 64ig_2^3\, N_c^{-2}\, G^>(x_3,z_1)\; G^>(z_3,z_1)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_2,z_2)\; G_A(z_3,z_3)\; G_K(x_1,z_1)\; G_R(z_2,z_1)$$

$$\mathcal{T}_{25,14} = 64ig_2^3\, N_c^{-2}\, G^>(x_3,z_1)\; G^>(z_3,z_1)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_2,z_2)\; G_A(z_3,z_3)\; G_K(z_2,z_1)\; G_R(x_1,z_1)$$

$$\mathcal{T}_{25,15} = 64ig_2^3\, N_c^{-2}\, G^>(x_3,z_1)\; G^>(z_3,z_1)\; G_A(x_2,z_2)\; G_A(x_4,z_3)\; G_A(z_2,z_2)\; G_K(x_1,z_1)\; G_R(z_2,z_1)\; G_R(z_3,z_3)$$

$\mathcal{T}_{25,16} = 64ig_2^3\, N_c^{-2}\, G^{>}(x_3, z_1)\; G^{>}(z_3, z_1)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_A(z_2, z_2)\; G_K(z_2, z_1)\; G_R(x_1, z_1)\; G_R(z_3, z_3)$

$\mathcal{T}_{25,17} = 64ig_2^3\, N_c^{-2}\, G^{>}(x_3, z_1)\; G^{>}(z_3, z_1)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_A(z_3, z_3)\; G_K(x_1, z_1)\; G_R(z_2, z_1)\; G_R(z_2, z_2)$

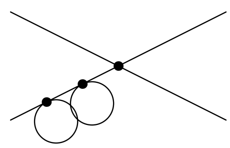
$\mathcal{T}_{25,18} = 64ig_2^3\, N_c^{-2}\, G^{>}(x_3, z_1)\; G^{>}(z_3, z_1)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_A(z_3, z_3)\; G_K(z_2, z_1)\; G_R(x_1, z_1)\; G_R(z_2, z_2)$

$\mathcal{T}_{25,19} = 64ig_2^3\, N_c^{-2}\, G^{>}(x_3, z_1)\; G^{>}(z_3, z_1)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_K(x_1, z_1)\; G_R(z_2, z_1)\; G_R(z_2, z_2)\; G_R(z_3, z_3)$

$\mathcal{T}_{25,20} = 64ig_2^3\, N_c^{-2}\, G^{>}(x_3, z_1)\; G^{>}(z_3, z_1)\; G_A(x_2, z_2)\; G_A(x_4, z_3)\; G_K(z_2, z_1)\; G_R(x_1, z_1)\; G_R(z_2, z_2)\; G_R(z_3, z_3)$

Topology 26 (FeynArts Topology 4)

2 → 2



1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 2. Net power in $C(t)$: -2. Explicit surviving terms:

$\mathcal{T}_{26,1} = 64ig_2^3\, N_c^{-2}\, G^{<}(x_1, z_1)\; G^{<}(z_3, z_1)\; G_A(x_2, z_2)\; G_A(x_4, z_1)\; G_A(z_2, z_2)\; G_A(z_3, z_2)\; G_K(z_3, z_3)\; G_R(x_3, z_1)$

$\mathcal{T}_{26,2} = 64ig_2^3\, N_c^{-2}\, G^{<}(x_1, z_1)\; G^{<}(z_3, z_1)\; G_A(x_2, z_2)\; G_A(x_4, z_1)\; G_A(z_2, z_2)\; G_A(z_3, z_3)\; G_K(z_3, z_2)\; G_R(x_3, z_1)$

$\mathcal{T}_{26,3} = 64ig_2^3\, N_c^{-2}\, G^{<}(x_1, z_1)\; G^{<}(z_3, z_1)\; G_A(x_2, z_2)\; G_A(x_4, z_1)\; G_A(z_2, z_2)\; G_K(z_3, z_2)\; G_R(x_3, z_1)\; G_R(z_3, z_3)$

$\mathcal{T}_{26,4} = 64ig_2^3\, N_c^{-2}\, G^{<}(x_1, z_1)\; G^{<}(z_3, z_1)\; G_A(x_2, z_2)\; G_A(x_4, z_1)\; G_A(z_3, z_2)\; G_K(z_3, z_3)\; G_R(x_3, z_1)\; G_R(z_2, z_2)$

$\mathcal{T}_{26,5} = 64ig_2^3\, N_c^{-2}\, G^{<}(x_1, z_1)\; G^{<}(z_3, z_1)\; G_A(x_2, z_2)\; G_A(x_4, z_1)\; G_A(z_3, z_3)\; G_K(z_2, z_2)\; G_R(x_3, z_1)\; G_R(z_3, z_2)$

$\mathcal{T}_{26,6} = 64ig_2^3\, N_c^{-2}\, G^{<}(x_1, z_1)\; G^{<}(z_3, z_1)\; G_A(x_2, z_2)\; G_A(x_4, z_1)\; G_A(z_3, z_3)\; G_K(z_3, z_2)\; G_R(x_3, z_1)\; G_R(z_2, z_2)$

$\mathcal{T}_{26,7} = 64ig_2^3\, N_c^{-2}\, G^{<}(x_1, z_1)\; G^{<}(z_3, z_1)\; G_A(x_2, z_2)\; G_A(x_4, z_1)\; G_K(z_2, z_2)\; G_R(x_3, z_1)\; G_R(z_3, z_2)\; G_R(z_3, z_3)$

$\mathcal{T}_{26,8} = 64ig_2^3\, N_c^{-2}\, G^{<}(x_1, z_1)\; G^{<}(z_3, z_1)\; G_A(x_2, z_2)\; G_A(x_4, z_1)\; G_K(z_3, z_2)\; G_R(x_3, z_1)\; G_R(z_2, z_2)\; G_R(z_3, z_3)$

$\mathcal{T}_{26,9} = 64ig_2^3\, N_c^{-2}\, G^{<}(x_1, z_1)\; G^{>}(z_3, z_2)\; G_A(x_2, z_2)\; G_A(x_4, z_1)\; G_A(z_2, z_2)\; G_A(z_3, z_1)\; G_K(z_3, z_3)\; G_R(x_3, z_1)$

$\mathcal{T}_{26,10} = 64ig_2^3\, N_c^{-2}\, G^{<}(x_1, z_1)\; G^{>}(z_3, z_2)\; G_A(x_2, z_2)\; G_A(x_4, z_1)\; G_A(z_2, z_2)\; G_A(z_3, z_3)\; G_K(x_3, z_1)\; G_R(z_3, z_1)$

$\mathcal{T}_{26,11} = 64ig_2^3\, N_c^{-2}\, G^{<}(x_1, z_1)\; G^{>}(z_3, z_2)\; G_A(x_2, z_2)\; G_A(x_4, z_1)\; G_A(z_2, z_2)\; G_A(z_3, z_3)\; G_K(z_3, z_1)\; G_R(x_3, z_1)$

$\mathcal{T}_{26,12} = 64ig_2^3\, N_c^{-2}\, G^{<}(x_1, z_1)\; G^{>}(z_3, z_2)\; G_A(x_2, z_2)\; G_A(x_4, z_1)\; G_A(z_2, z_2)\; G_K(x_3, z_1)\; G_R(z_3, z_1)\; G_R(z_3, z_3)$

$\mathcal{T}_{26,13} = 64ig_2^3\, N_c^{-2}\, G^{<}(x_1, z_1)\; G^{>}(z_3, z_2)\; G_A(x_2, z_2)\; G_A(x_4, z_1)\; G_A(z_2, z_2)\; G_K(z_3, z_1)\; G_R(x_3, z_1)\; G_R(z_3, z_3)$

$\mathcal{T}_{26,14} = 64ig_2^3\, N_c^{-2}\, G^{<}(x_1, z_1)\; G^{>}(z_3, z_2)\; G_A(x_2, z_2)\; G_A(x_4, z_1)\; G_A(z_3, z_1)\; G_K(z_3, z_3)\; G_R(x_3, z_1)\; G_R(z_2, z_2)$

$\mathcal{T}_{26,15} = 64ig_2^3\, N_c^{-2}\, G^{<}(x_1, z_1)\; G^{>}(z_3, z_2)\; G_A(x_2, z_2)\; G_A(x_4, z_1)\; G_A(z_3, z_3)\; G_K(x_3, z_1)\; G_R(z_2, z_2)\; G_R(z_3, z_1)$

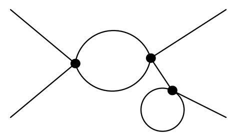
$\mathcal{T}_{26,16} = 64ig_2^3\, N_c^{-2}\, G^{<}(x_1, z_1)\; G^{>}(z_3, z_2)\; G_A(x_2, z_2)\; G_A(x_4, z_1)\; G_A(z_3, z_3)\; G_K(z_3, z_1)\; G_R(x_3, z_1)\; G_R(z_2, z_2)$

$\mathcal{T}_{26,17} = 64ig_2^3\, N_c^{-2}\, G^{<}(x_1, z_1)\; G^{>}(z_3, z_2)\; G_A(x_2, z_2)\; G_A(x_4, z_1)\; G_K(x_3, z_1)\; G_R(z_2, z_2)\; G_R(z_3, z_1)\; G_R(z_3, z_3)$

$\mathcal{T}_{26,18} = 64ig_2^3\, N_c^{-2}\, G^{<}(x_1, z_1)\; G^{>}(z_3, z_2)\; G_A(x_2, z_2)\; G_A(x_4, z_1)\; G_K(z_3, z_1)\; G_R(x_3, z_1)\; G_R(z_2, z_2)\; G_R(z_3, z_3)$

Topology 27 (FeynArts Topology 4)

2 $\rightarrow$ 2



1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\mathcal{T}_{27,1} = 128ig_2^3\, N_c^{-3}\, G^>(x_3,z_2)\; G^>(z_3,z_2)\; G_A(x_2,z_1)\; G_A(x_4,z_3)\; G_A(z_2,z_1)\; G_A(z_3,z_3)\; G_K(x_1,z_1)\; G_R(z_2,z_1)$$

$$\mathcal{T}_{27,2} = 128ig_2^3\, N_c^{-3}\, G^>(x_3,z_2)\; G^>(z_3,z_2)\; G_A(x_2,z_1)\; G_A(x_4,z_3)\; G_A(z_2,z_1)\; G_A(z_3,z_3)\; G_K(z_2,z_1)\; G_R(x_1,z_1)$$

$$\mathcal{T}_{27,3} = 128ig_2^3\, N_c^{-3}\, G^>(x_3,z_2)\; G^>(z_3,z_2)\; G_A(x_2,z_1)\; G_A(x_4,z_3)\; G_A(z_2,z_1)\; G_K(x_1,z_1)\; G_R(z_2,z_1)\; G_R(z_3,z_3)$$

$$\mathcal{T}_{27,4} = 128ig_2^3\, N_c^{-3}\, G^>(x_3,z_2)\; G^>(z_3,z_2)\; G_A(x_2,z_1)\; G_A(x_4,z_3)\; G_A(z_2,z_1)\; G_K(z_2,z_1)\; G_R(x_1,z_1)\; G_R(z_3,z_3)$$

$$\mathcal{T}_{27,5} = 64ig_2^3\, N_c^{-3}\, G^>(z_2,z_1)\; G^>(z_2,z_1)\; G_A(x_2,z_1)\; G_A(x_4,z_3)\; G_A(z_3,z_2)\; G_K(z_3,z_3)\; G_R(x_1,z_1)\; G_R(x_3,z_2)$$

$$\mathcal{T}_{27,6} = 64ig_2^3\, N_c^{-3}\, G^>(z_2,z_1)\; G^>(z_2,z_1)\; G_A(x_2,z_1)\; G_A(x_4,z_3)\; G_A(z_3,z_3)\; G_K(x_3,z_2)\; G_R(x_1,z_1)\; G_R(z_3,z_2)$$

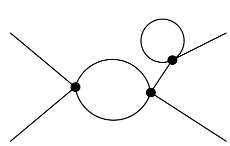
$$\mathcal{T}_{27,7} = 64ig_2^3\, N_c^{-3}\, G^>(z_2,z_1)\; G^>(z_2,z_1)\; G_A(x_2,z_1)\; G_A(x_4,z_3)\; G_A(z_3,z_3)\; G_K(z_3,z_2)\; G_R(x_1,z_1)\; G_R(x_3,z_2)$$

$$\mathcal{T}_{27,8} = 64ig_2^3\, N_c^{-3}\, G^>(z_2,z_1)\; G^>(z_2,z_1)\; G_A(x_2,z_1)\; G_A(x_4,z_3)\; G_K(x_3,z_2)\; G_R(x_1,z_1)\; G_R(z_3,z_2)\; G_R(z_3,z_3)$$

$$\mathcal{T}_{27,9} = 64ig_2^3\, N_c^{-3}\, G^>(z_2,z_1)\; G^>(z_2,z_1)\; G_A(x_2,z_1)\; G_A(x_4,z_3)\; G_K(z_3,z_2)\; G_R(x_1,z_1)\; G_R(x_3,z_2)\; G_R(z_3,z_3)$$

Topology 28 (FeynArts Topology 4)

2 $\rightarrow$ 2



1

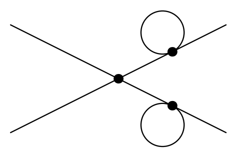
Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\mathcal{T}_{28,1} = 64ig_2^3\, N_c^{-3}\, G^{>}(z_3, z_1)\, G^{>}(z_3, z_1)\, G_A(x_2, z_1)\, G_A(x_4, z_3)\, G_A(z_2, z_2)\, G_A(z_3, z_2)\, G_K(x_3, z_2)\, G_R(x_1, z_1)$$

$$\mathcal{T}_{28,2} = 64ig_2^3\, N_c^{-3}\, G^{>}(z_3, z_1)\, G^{>}(z_3, z_1)\, G_A(x_2, z_1)\, G_A(x_4, z_3)\, G_A(z_3, z_2)\, G_K(x_3, z_2)\, G_R(x_1, z_1)\, G_R(z_2, z_2)$$

$$\mathcal{T}_{28,3} = 64ig_2^3\, N_c^{-3}\, G^{>}(z_3, z_1)\, G^{>}(z_3, z_1)\, G_A(x_2, z_1)\, G_A(x_4, z_3)\, G_A(z_3, z_2)\, G_K(z_2, z_2)\, G_R(x_1, z_1)\, G_R(x_3, z_2)$$

Topology 29 (FeynArts Topology 4)



Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 2. Net power in $C(t)$: -2. Explicit surviving terms:

$$\mathcal{T}_{29,1} = 64ig_2^3\, N_c^{-2}\, G^{>}(x_3, z_2)\, G^{>}(z_3, z_1)\, G_A(x_2, z_1)\, G_A(x_4, z_3)\, G_A(z_2, z_1)\, G_A(z_3, z_3)\, G_K(z_2, z_2)\, G_R(x_1, z_1)$$

$$\mathcal{T}_{29,2} = 64ig_2^3\, N_c^{-2}\, G^{>}(x_3, z_2)\, G^{>}(z_3, z_1)\, G_A(x_2, z_1)\, G_A(x_4, z_3)\, G_A(z_2, z_1)\, G_K(z_2, z_2)\, G_R(x_1, z_1)\, G_R(z_3, z_3)$$

$$\mathcal{T}_{29,3} = 64ig_2^3\, N_c^{-2}\, G^{>}(x_3, z_2)\, G^{>}(z_3, z_1)\, G_A(x_2, z_1)\, G_A(x_4, z_3)\, G_A(z_2, z_2)\, G_A(z_3, z_3)\, G_K(x_1, z_1)\, G_R(z_2, z_1)$$

$$\mathcal{T}_{29,4} = 64ig_2^3\, N_c^{-2}\, G^{>}(x_3, z_2)\, G^{>}(z_3, z_1)\, G_A(x_2, z_1)\, G_A(x_4, z_3)\, G_A(z_2, z_2)\, G_A(z_3, z_3)\, G_K(z_2, z_1)\, G_R(x_1, z_1)$$

$$\mathcal{T}_{29,5} = 64ig_2^3\, N_c^{-2}\, G^{>}(x_3, z_2)\, G^{>}(z_3, z_1)\, G_A(x_2, z_1)\, G_A(x_4, z_3)\, G_A(z_2, z_2)\, G_K(x_1, z_1)\, G_R(z_2, z_1)\, G_R(z_3, z_3)$$

$$\mathcal{T}_{29,6} = 64ig_2^3\, N_c^{-2}\, G^{>}(x_3, z_2)\, G^{>}(z_3, z_1)\, G_A(x_2, z_1)\, G_A(x_4, z_3)\, G_A(z_2, z_2)\, G_K(z_2, z_1)\, G_R(x_1, z_1)\, G_R(z_3, z_3)$$

$$\mathcal{T}_{29,7} = 64ig_2^3\, N_c^{-2}\, G^{>}(x_3, z_2)\, G^{>}(z_3, z_1)\, G_A(x_2, z_1)\, G_A(x_4, z_3)\, G_A(z_3, z_3)\, G_K(x_1, z_1)\, G_R(z_2, z_1)\, G_R(z_2, z_2)$$

$$\mathcal{T}_{29,8} = 64ig_2^3\, N_c^{-2}\, G^{>}(x_3, z_2)\, G^{>}(z_3, z_1)\, G_A(x_2, z_1)\, G_A(x_4, z_3)\, G_A(z_3, z_3)\, G_K(z_2, z_1)\, G_R(x_1, z_1)\, G_R(z_2, z_2)$$

$$\mathcal{T}_{29,9} = 64ig_2^3\, N_c^{-2}\, G^{>}(x_3, z_2)\, G^{>}(z_3, z_1)\, G_A(x_2, z_1)\, G_A(x_4, z_3)\, G_K(x_1, z_1)\, G_R(z_2, z_1)\, G_R(z_2, z_2)\, G_R(z_3, z_3)$$

$$\mathcal{T}_{29,10} = 64ig_2^3\, N_c^{-2}\, G^{>}(x_3, z_2)\, G^{>}(z_3, z_1)\, G_A(x_2, z_1)\, G_A(x_4, z_3)\, G_K(z_2, z_1)\, G_R(x_1, z_1)\, G_R(z_2, z_2)\, G_R(z_3, z_3)$$

$$\mathcal{T}_{29,11} = 64ig_2^3\, N_c^{-2}\, G^{>}(z_2, z_1)\, G^{>}(z_3, z_1)\, G_A(x_2, z_1)\, G_A(x_4, z_3)\, G_A(z_2, z_2)\, G_A(z_3, z_3)\, G_K(x_3, z_2)\, G_R(x_1, z_1)$$

$$\mathcal{T}_{29,12} = 64ig_2^3\, N_c^{-2}\, G^{>}(z_2, z_1)\, G^{>}(z_3, z_1)\, G_A(x_2, z_1)\, G_A(x_4, z_3)\, G_A(z_2, z_2)\, G_K(x_3, z_2)\, G_R(x_1, z_1)\, G_R(z_3, z_3)$$

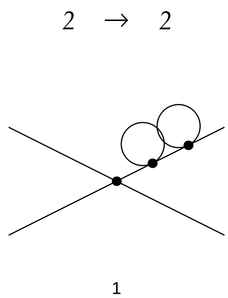
$$\mathcal{T}_{29,13} = 64ig_2^3\, N_c^{-2}\, G^{>}(z_2, z_1)\, G^{>}(z_3, z_1)\, G_A(x_2, z_1)\, G_A(x_4, z_3)\, G_A(z_3, z_3)\, G_K(x_3, z_2)\, G_R(x_1, z_1)\, G_R(z_2, z_2)$$

$$\mathcal{T}_{29,14} = 64ig_2^3\, N_c^{-2}\, G^{>}(z_2, z_1)\, G^{>}(z_3, z_1)\, G_A(x_2, z_1)\, G_A(x_4, z_3)\, G_A(z_3, z_3)\, G_K(z_2, z_2)\, G_R(x_1, z_1)\, G_R(x_3, z_2)$$

$$\mathcal{T}_{29,15} = 64ig_2^3\, N_c^{-2}\, G^{>}(z_2, z_1)\, G^{>}(z_3, z_1)\, G_A(x_2, z_1)\, G_A(x_4, z_3)\, G_K(x_3, z_2)\, G_R(x_1, z_1)\, G_R(z_2, z_2)\, G_R(z_3, z_3)$$

$$\mathcal{T}_{29,16} = 64ig_2^3\, N_c^{-2}\, G^{>}(z_2, z_1)\, G^{>}(z_3, z_1)\, G_A(x_2, z_1)\, G_A(x_4, z_3)\, G_K(z_2, z_2)\, G_R(x_1, z_1)\, G_R(x_3, z_2)\, G_R(z_3, z_3)$$

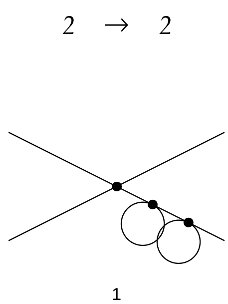
Topology 30 (FeynArts Topology 4)



Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 2. Net power in $C(t)$: -2.

No surviving SK contributions for this topology.

Topology 31 (FeynArts Topology 4)



Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 2. Net power in $C(t)$: -2. Explicit surviving terms:

$$\mathcal{T}_{31,1} = 64ig_2^3\, N_c^{-2}\, G^{<}(z_3,z_2)\; G^{>}(x_3,z_1)\; G_A(x_2,z_1)\; G_A(x_4,z_2)\; G_A(z_2,z_2)\; G_A(z_3,z_1)\; G_K(z_3,z_3)\; G_R(x_1,z_1)$$

$$\mathcal{T}_{31,2} = 64ig_2^3\, N_c^{-2}\, G^{<}(z_3,z_2)\; G^{>}(x_3,z_1)\; G_A(x_2,z_1)\; G_A(x_4,z_2)\; G_A(z_2,z_2)\; G_A(z_3,z_3)\; G_K(x_1,z_1)\; G_R(z_3,z_1)$$

$$\mathcal{T}_{31,3} = 64ig_2^3\, N_c^{-2}\, G^{<}(z_3,z_2)\; G^{>}(x_3,z_1)\; G_A(x_2,z_1)\; G_A(x_4,z_2)\; G_A(z_2,z_2)\; G_A(z_3,z_3)\; G_K(z_3,z_1)\; G_R(x_1,z_1)$$

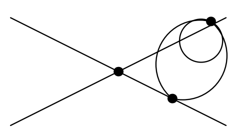
$$\mathcal{T}_{31,4} = 64ig_2^3\, N_c^{-2}\, G^{<}(z_3,z_2)\; G^{>}(x_3,z_1)\; G_A(x_2,z_1)\; G_A(x_4,z_2)\; G_A(z_2,z_2)\; G_K(x_1,z_1)\; G_R(z_3,z_1)\; G_R(z_3,z_3)$$

$$\mathcal{T}_{31,5} = 64ig_2^3\, N_c^{-2}\, G^{<}(z_3,z_2)\; G^{>}(x_3,z_1)\; G_A(x_2,z_1)\; G_A(x_4,z_2)\; G_A(z_2,z_2)\; G_K(z_3,z_1)\; G_R(x_1,z_1)\; G_R(z_3,z_3)$$

$$\mathcal{T}_{31,6} = 64ig_2^3\,N_c^{-2}\,G^{<}(z_3,z_2)\;G^{>}(x_3,z_1)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_A(z_3,z_1)\;G_K(z_3,z_3)\;G_R(x_1,z_1)\;G_R(z_2,z_2)$$
$$\mathcal{T}_{31,7} = 64ig_2^3\,N_c^{-2}\,G^{<}(z_3,z_2)\;G^{>}(x_3,z_1)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_A(z_3,z_3)\;G_K(x_1,z_1)\;G_R(z_2,z_2)\;G_R(z_3,z_1)$$
$$\mathcal{T}_{31,8} = 64ig_2^3\,N_c^{-2}\,G^{<}(z_3,z_2)\;G^{>}(x_3,z_1)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_A(z_3,z_3)\;G_K(z_3,z_1)\;G_R(x_1,z_1)\;G_R(z_2,z_2)$$
$$\mathcal{T}_{31,9} = 64ig_2^3\,N_c^{-2}\,G^{<}(z_3,z_2)\;G^{>}(x_3,z_1)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_K(x_1,z_1)\;G_R(z_2,z_2)\;G_R(z_3,z_1)\;G_R(z_3,z_3)$$
$$\mathcal{T}_{31,10} = 64ig_2^3\,N_c^{-2}\,G^{<}(z_3,z_2)\;G^{>}(x_3,z_1)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_K(z_3,z_1)\;G_R(x_1,z_1)\;G_R(z_2,z_2)\;G_R(z_3,z_3)$$
$$\mathcal{T}_{31,11} = 64ig_2^3\,N_c^{-2}\,G^{>}(x_3,z_1)\;G^{>}(z_3,z_1)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_A(z_2,z_2)\;G_A(z_3,z_2)\;G_K(z_3,z_3)\;G_R(x_1,z_1)$$
$$\mathcal{T}_{31,12} = 64ig_2^3\,N_c^{-2}\,G^{>}(x_3,z_1)\;G^{>}(z_3,z_1)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_A(z_2,z_2)\;G_A(z_3,z_3)\;G_K(z_3,z_2)\;G_R(x_1,z_1)$$
$$\mathcal{T}_{31,13} = 64ig_2^3\,N_c^{-2}\,G^{>}(x_3,z_1)\;G^{>}(z_3,z_1)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_A(z_2,z_2)\;G_K(z_3,z_2)\;G_R(x_1,z_1)\;G_R(z_3,z_3)$$
$$\mathcal{T}_{31,14} = 64ig_2^3\,N_c^{-2}\,G^{>}(x_3,z_1)\;G^{>}(z_3,z_1)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_A(z_3,z_2)\;G_K(z_3,z_3)\;G_R(x_1,z_1)\;G_R(z_2,z_2)$$
$$\mathcal{T}_{31,15} = 64ig_2^3\,N_c^{-2}\,G^{>}(x_3,z_1)\;G^{>}(z_3,z_1)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_A(z_3,z_3)\;G_K(z_2,z_2)\;G_R(x_1,z_1)\;G_R(z_3,z_2)$$
$$\mathcal{T}_{31,16} = 64ig_2^3\,N_c^{-2}\,G^{>}(x_3,z_1)\;G^{>}(z_3,z_1)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_A(z_3,z_3)\;G_K(z_3,z_2)\;G_R(x_1,z_1)\;G_R(z_2,z_2)$$
$$\mathcal{T}_{31,17} = 64ig_2^3\,N_c^{-2}\,G^{>}(x_3,z_1)\;G^{>}(z_3,z_1)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_K(z_2,z_2)\;G_R(x_1,z_1)\;G_R(z_3,z_2)\;G_R(z_3,z_3)$$
$$\mathcal{T}_{31,18} = 64ig_2^3\,N_c^{-2}\,G^{>}(x_3,z_1)\;G^{>}(z_3,z_1)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_K(z_3,z_2)\;G_R(x_1,z_1)\;G_R(z_2,z_2)\;G_R(z_3,z_3)$$

Topology 32 (FeynArts Topology 4)

2 → 2



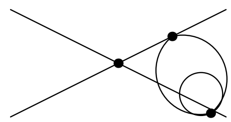
1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 2. Net power in $C(t)$: -2. Explicit surviving terms:

$$\mathcal{T}_{32,1} = 128ig_2^3\,N_c^{-2}\,G^{<}(z_1,z_2)\;G^{>}(x_3,z_1)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_A(z_2,z_3)\;G_A(z_3,z_3)\;G_K(z_2,z_3)\;G_R(x_1,z_1)$$
$$\mathcal{T}_{32,2} = 128ig_2^3\,N_c^{-2}\,G^{<}(z_1,z_2)\;G^{>}(x_3,z_1)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_A(z_2,z_3)\;G_K(z_2,z_3)\;G_R(x_1,z_1)\;G_R(z_3,z_3)$$
$$\mathcal{T}_{32,3} = 128ig_2^3\,N_c^{-2}\,G^{<}(z_1,z_2)\;G^{>}(x_3,z_1)\;G_A(x_2,z_1)\;G_A(x_4,z_2)\;G_A(z_2,z_3)\;G_K(z_3,z_3)\;G_R(x_1,z_1)\;G_R(z_2,z_3)$$

Topology 33 (FeynArts Topology 4)

$2 \quad \to \quad 2$



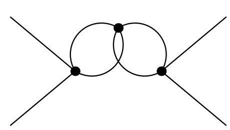
1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 2. Net power in $C(t)$: -2.

No surviving SK contributions for this topology.

Topology 34 (FeynArts Topology 4)

$2 \quad \to \quad 2$



1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3. Explicit surviving terms:

$$\mathcal{T}_{34,1} = 128 i g_2^3 \, N_c^{-3} \, G^{<}(z_1, z_3) \; G^{<}(z_1, z_3) \; G_A(x_2, z_1) \; G_A(x_4, z_2) \; G_A(z_2, z_3) \; G_K(x_3, z_2) \; G_R(x_1, z_1) \; G_R(z_2, z_3)$$

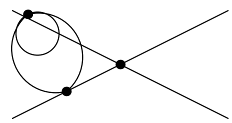
$$\mathcal{T}_{34,2} = 128 i g_2^3 \, N_c^{-3} \, G^{<}(z_1, z_3) \; G^{<}(z_1, z_3) \; G_A(x_2, z_1) \; G_A(x_4, z_2) \; G_K(z_2, z_3) \; G_R(x_1, z_1) \; G_R(x_3, z_2) \; G_R(z_2, z_3)$$

$$\mathcal{T}_{34,3} = 128 i g_2^3 \, N_c^{-3} \, G^{>}(z_2, z_3) \; G^{>}(z_2, z_3) \; G_A(x_2, z_1) \; G_A(x_4, z_2) \; G_A(z_1, z_3) \; G_K(x_1, z_1) \; G_R(x_3, z_2) \; G_R(z_1, z_3)$$

$$\mathcal{T}_{34,4} = 128 i g_2^3 \, N_c^{-3} \, G^{>}(z_2, z_3) \; G^{>}(z_2, z_3) \; G_A(x_2, z_1) \; G_A(x_4, z_2) \; G_K(z_1, z_3) \; G_R(x_1, z_1) \; G_R(x_3, z_2) \; G_R(z_1, z_3)$$

Topology 35 (FeynArts Topology 4)

2 → 2



1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 2. Net power in $C(t)$: -2. Explicit surviving terms:

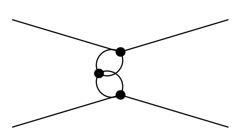
$$\mathcal{T}_{35,1} = 128ig_2^3\,N_c^{-2}\,G^{<}\left(x_1,z_1\right)\,G^{>}\left(z_1,z_2\right)\,G_A\left(x_2,z_2\right)\,G_A\left(x_4,z_1\right)\,G_A\left(z_2,z_3\right)\,G_A\left(z_3,z_3\right)\,G_K\left(z_2,z_3\right)\,G_R\left(x_3,z_1\right)$$

$$\mathcal{T}_{35,2} = 128ig_2^3\,N_c^{-2}\,G^{<}\left(x_1,z_1\right)\,G^{>}\left(z_1,z_2\right)\,G_A\left(x_2,z_2\right)\,G_A\left(x_4,z_1\right)\,G_A\left(z_2,z_3\right)\,G_K\left(z_2,z_3\right)\,G_R\left(x_3,z_1\right)\,G_R\left(z_3,z_3\right)$$

$$\mathcal{T}_{35,3} = 128ig_2^3\,N_c^{-2}\,G^{<}\left(x_1,z_1\right)\,G^{>}\left(z_1,z_2\right)\,G_A\left(x_2,z_2\right)\,G_A\left(x_4,z_1\right)\,G_A\left(z_2,z_3\right)\,G_K\left(z_3,z_3\right)\,G_R\left(x_3,z_1\right)\,G_R\left(z_2,z_3\right)$$

Topology 36 (FeynArts Topology 4)

2 → 2

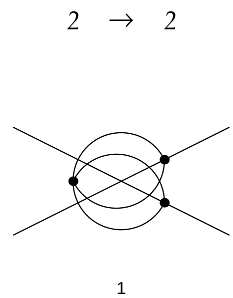


1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3.

No surviving SK contributions for this topology.

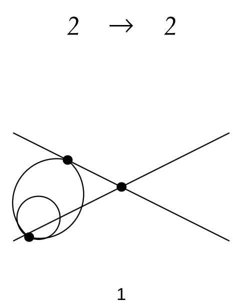
Topology 37 (FeynArts Topology 4)



Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 1. Net power in $C(t)$: -3.

No surviving SK contributions for this topology.

Topology 38 (FeynArts Topology 4)

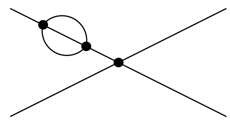


Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 2. Net power in $C(t)$: -2.

No surviving SK contributions for this topology.

Topology 39 (FeynArts Topology 6)

2 → 2



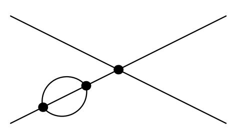
1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 0. Net power in $C(t)$: -4.

No surviving SK contributions for this topology.

Topology 40 (FeynArts Topology 6)

2 → 2



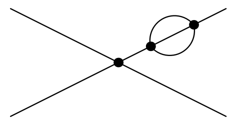
1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 0. Net power in $C(t)$: -4. Explicit surviving terms:

$$\mathcal{T}_{40,1} = 384 i g_2^3 \, N_c^{-4} \, G^{<}(x_1, z_1) \; G^{<}(z_3, z_1) \; G_A(x_2, z_2) \; G_A(x_4, z_1) \; G_A(z_3, z_2) \; G_K(z_3, z_2) \; G_R(x_3, z_1) \; G_R(z_3, z_2)$$

Topology 41 (FeynArts Topology 6)

$2 \rightarrow 2$



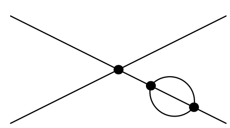
1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 0. Net power in $C(t)$: -4.

No surviving SK contributions for this topology.

Topology 42 (FeynArts Topology 6)

$2 \rightarrow 2$



1

Parsed propagators: 8. Internal vertices: 3. Internal half-edges: 12.
Assignments checked: 32768. Surviving (nonzero vertex weight): 4096.
Raw N_c power: 0. Net power in $C(t)$: -4. Explicit surviving terms:

$$\mathcal{T}_{42,1} = 384 i g_2^3 \, N_c^{-4} \, G^{>}(x_3, z_1) \, G^{>}(z_3, z_1) \, G_A(x_2, z_1) \, G_A(x_4, z_2) \, G_A(z_3, z_2) \, G_K(z_3, z_2) \, G_R(x_1, z_1) \, G_R(z_3, z_2)$$